SYLLABUS : SEMICONDUCTOR

1. The barrier potential in a P-N junction is maximum in-

(A) the reverse bias condition

(B) the forward bias condition

(C) the condition when the junction diode is used as rectifier

(D) zero bias condition

2. The diffusion current in a P-N junction flows-

(A) from the N-side to the P-side

(B) from the P-side to the N-side

(C) from the N-side to the P-side if the junction is forward-biased and in the opposite direction if it is reverse baised

(D) from the P-side to the N-side if the junction is forward-biased and in the opposite direction if it is reverse biased

3. The value of barrier potential of P-N junction in Ge is-

(A) 0.03 volt in the direction of forward current

(B) 0.3 volt in the direction opposite of the forward current

(C) 25 volt in the direction opposite to the forward current

(D) 25 volt in the direction of the forward current

4. Region which have no free electron and holes in a P-N juction is-

(A) P-region (B) N-region (C) junction (D) depletion region

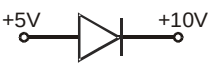
5. The P-N junction diode works as insulator, if connected-

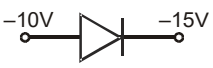
(A) To a.c. (B) In forward bias (C) In reverse bias (D) None of these

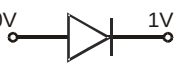
6. If the forward voltage in a diode is increased, the width of the depletion region-

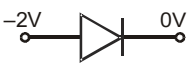
(A) Decreases (B) Increases (C) Fluctuates (D) does not change

7. In which case is the juction diode is not reverse bias-

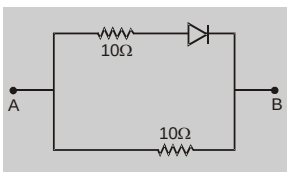
(A) 

(B) 

(C) 

(D) 

8. If V_A and V_B denote the potentials of A and B, then the equivalent resistance between A and B in the adjoint electric circuit is-



(A) 10 ohm if $V_A > V_B$ (B) 5 ohm if $V_A < V_B$ (C) 5 ohm if $V_A > V_B$ (D) 20 ohm if $V_A > V_B$

9. If the two ends of a P-N junction are joined by a wire-

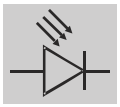
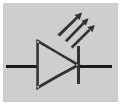
(A) There will not be a steady current in the circuit

(B) There will be a steady current from the N-side to the P-side

(C) There will a steady current from the P-side to the N-side

(D) There may or may not be a current depending upon the resistance of the connecting wire.

10. Symbolic representation of photodiode is-

(A)  (B)  (C)  (D) 

11. Consider the following statements 1 and 2 and identify the correct answer-

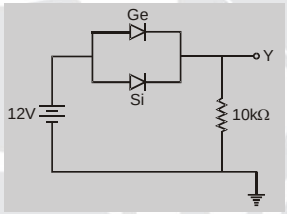
[1] A zener diode is always connected in reverse bias.

[2] The potential barrier of a P-N junction lies between 0.1 to 0.3 V approximately.

(A) [1] and [2] are correct (B) [1] and [2] are wrong

(C) [1] is correct, but [2] is wrong (D) [1] is wrong, but [2] is correct

12. Two junction diodes one of germanium (Ge) and other of silicon (Si) are connected as shown in figure to a battery of emf 12 V and a load resistance 10 kΩ. The germanium diode conducts at 0.3 V and silicon diode at 0.7 V. When a current flows in the circuit, the potential of terminal Y will be-



(A) 12 V (B) 11 V (C) 11.3 V (D) 11.7 V

13. When the reverse potential in a semiconductor diode are 10V and 20V, then the corresponding reverse currents are 25μA and 50μA respectively. The reverse resistance of junction diode will be-

(A) 40Ω (B) 4×10^5 Ω (C) 40KΩ (D) 4×10^{-5} Ω

14. If the frequency of input alternating potential is n, then the ripple frequency of output potential of full wave rectifier will be-

(A) 2n (B) n (C) $\frac{n}{2}$ (D) $\frac{n}{4}$

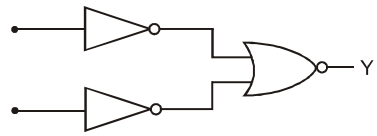
15. Function of rectifier is

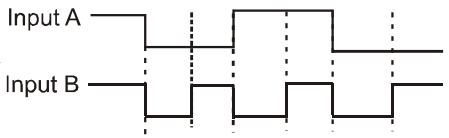
(A) To convert ac into dc (B) To convert dc into ac

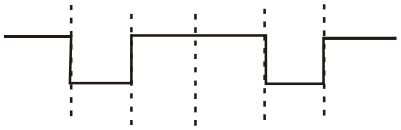
(C) Both (1) and (2) (D) None of these

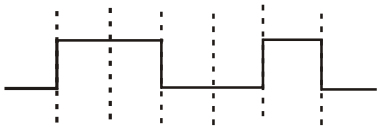
Power of real gurus

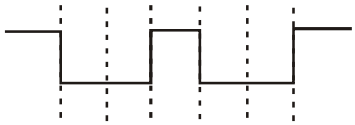
16. The logic circuit shown below has the input waveforms 'A' and 'B' as shown. Pick out the correct out put waveform.

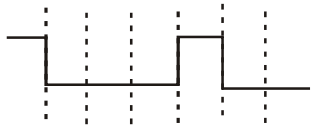




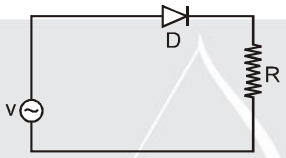
(A) 

(B) 

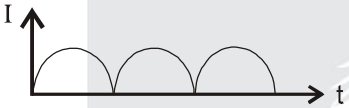
(C) 

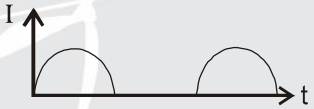
(D) 

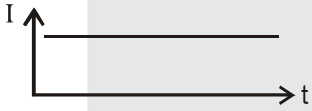
17. A p-n junction (D) shown in the figure can act as a rectifier. An alternating current source (V) is connected in the circuit.



The current (I) in the resistor (R) can be shown by:

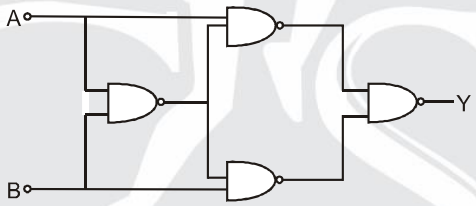
(A) 

(B) 

(C) 

(D) 

18. Truth table for system of four NAND gates as shown in figure is :



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

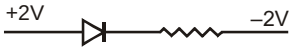
(C)

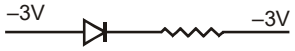
A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

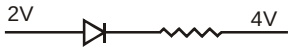
(D)

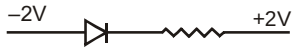
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

19. The forward biased diode connection is

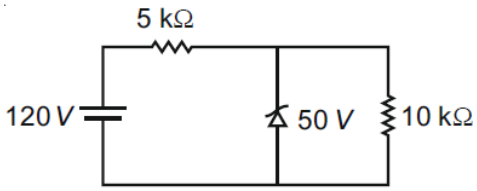
(A) 

(B) 

(C) 

(D) 

20. For the circuit shown below, the current through the Zener diode is:



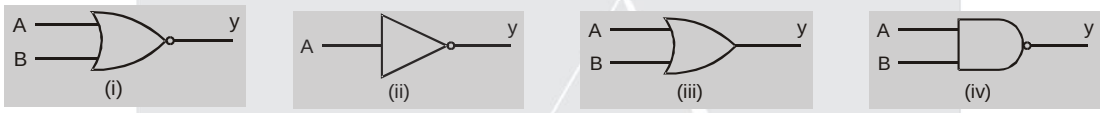
(A) Zero (B) 9 mA (C) 14 mA (D) 5 mA

21. The truth table given below is for :–

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

(A) OR gate (B) AND gate (C) XNOR gate (D) XOR gate

22. Given below are four logic gates symbols. NAND, NOR and OR are respectively-



(A) (iv), (i), (iii) (B) (i), (iii), (iv) (C) (i), (ii), (iii) (D) (i), (iv), (ii)

23. NAND gate is-

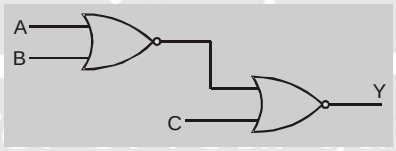
(A) a basic gate (B) not a universal gate (C) a universal gate (D) none of these.

24. The given truth table is for which logic gate-

A	B	Y
1	1	0
0	1	1
1	0	1
0	0	1

(A) NAND (B) XOR (C) NOR (D) OR

25. The Boolean expression for the output Y of the logic operation shown, is-



(A) $\overline{(A+B)}+C$ (B) $(A+B)+\overline{C}$ (C) $\overline{(A+B)+C}$ (D) $\overline{(A+B+C)}$

ANSWER KEY

1.	(A)	**2.**	(B)	**3.**	(B)	**4.**	(D)	**5.**	(C)
6.	(A)	**7.**	(B)	**8.**	(C)	**9.**	(A)	**10.**	(C)
11.	(C)	**12.**	(D)	**13.**	(B)	**14.**	(A)	**15.**	(A)
16.	(D)	**17.**	(B)	**18.**	(A)	**19.**	(A)	**20.**	(B)
21.	(C)	**22.**	(A)	**23.**	(C)	**24.**	(A)	**25.**	(C)